

Treewidth-Based Separation for SAT: Rail-Admissible Computation

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Abstract

We define a computational model called rail-admissible computation characterized by bounded local degrees of freedom ($d \leq 3$) and bounded separator size ($k \leq 12$) at module interfaces. We prove this model is equivalent to bounded-treewidth computation and establish that SAT requires unbounded treewidth via expander-based hard instances. This yields a clean separation: $\text{SAT} \notin \text{Rail-Admissible-Polytime}$. Whether this model equals P remains open and determines if this result implies $P \neq \text{NP}$.

1. The Rail-Admissible Model

A **primitive gate** is a finite graph with maximum degree ≤ 3 representing a basic computational operation.

A **module** M is built from primitive gates with a boundary $\partial M \subseteq V(M)$ of exposed vertices.

A **rail-join** of modules M_1, M_2 along separator S produces $M_1 \oplus_S M_2$ by identifying boundary vertices and making S a clique ($|S| \leq k$).

A **k-rail-admissible composition** is a family of modules $\{M_i\}$ with:

- Tree structure T on modules (no cyclic dependencies)
- Each edge has separator $|S_{ij}| \leq k$ (the thickness bound)
- Running intersection property (vertices appearing in multiple modules form connected subtree)

Definition: A computation is **k-rail-admissible** if expressible as such a composition. We focus on $k=12$.

Intuition: Local operations touch ≤ 3 variables; interfaces between computational blocks involve ≤ 12 simultaneous "threads of control." This captures natural locality constraints.

2. Main Results

Lemma 1 (Structure): Rail-Admissible $\Rightarrow$ Bounded Treewidth

Theorem: Let G be a k -rail-admissible computation graph. Then $\text{tw}(G) \leq \max(w_0, k-1)$ where w_0 is the maximum treewidth of any single module (constant).

Proof sketch: By induction on modules. Base: single module has $\text{tw} \leq w_0$. Step: adding module via separator S with $|S| \leq k$, apply clique-sum bound: $\text{tw}(G \oplus_s M) \leq \max(\text{tw}(G), \text{tw}(M), |S|-1) \leq \max(w_0, k-1)$. The tree-of-joins structure prevents grid-like blowup that would increase treewidth.

Corollary: 12-rail-admissible $\Rightarrow \text{tw} = O(1)$.

Lemma 2 (Hardness): SAT Requires Unbounded Treewidth

Theorem: There exists a polynomial-time constructible family $\{F_n\}$ of unsatisfiable 3-CNF formulas with:

Resolution width: $w_{\text{res}}(F_n) \geq cn$

Treewidth: $\text{tw}(G_{\text{pr}}(F_n)) \geq cn-1$

Construction: Tseitin contradictions on d -regular expanders:

Take expander graph H_n on n vertices (e.g., Ramanujan graphs)

Assign odd total charge: $\chi(v^*)=1$ for one vertex, $\chi(v)=0$ elsewhere

Variable x_e per edge $e \in E(H_n)$

Encode parity constraints: $\bigoplus_{e \ni v} x_e = \chi(v)$ at each vertex

Convert to 3-CNF via standard gadgets $\rightarrow$ formula F_n

Key properties:

F_n unsatisfiable (odd total charge violates parity)

Primal graph $G_{\text{pr}}(F_n)$ inherits expander structure

Known result (Ben-Sasson & Wigderson 2001): $w_{\text{res}}(F_n) \geq \Omega(n)$

Standard theorem: $w_{\text{res}}(F) \leq \text{tw}(G_{\text{pr}}(F)) + 1$

Therefore: $\text{tw}(G_{\text{pr}}(F_n)) \geq \Omega(n)$

Lemma 3 (Communication): Bounded Treewidth $\Rightarrow$ Large Communication

Theorem: For lifted family $\phi_{\text{G},q}$ on $\Theta(nq)$ variables, any width- k tree decomposition has a separator requiring $\Omega(nq/(k+1))$ bits to evaluate SAT correctly.

Proof sketch:

Standard DP on tree decomposition sends messages across separators
 Each message encodes partial satisfiability on separator variables
 Expander structure forces $\Omega(nq)$ independent parity constraints to cross any balanced separator
 Separator of size k can encode at most $O(k)$ parities via variable assignments
 Remaining $\Omega(nq) - O(k)$ constraints require explicit communication
 With $k = O(1)$ and $q = \Theta(n)$: $\Omega(n^2)$ bits required

Interpretation: Even with bounded-width decomposition, computational work scales super-polynomially for $k = O(1)$.

3. Integrated Result

Theorem A (Main Result): SAT $\notin$ Rail-Admissible-Polytime

Proof: Assume rail-admissible polytime evaluator exists for SAT.

By Lemma 1: must use treewidth $k = O(1)$
 By Lemma 3: requires $\Omega(n^2)$ communication on $\phi_{\{G,n\}}$
 By Lemma 2: family $\{F_n\}$ has $tw = \Omega(n)$, cannot fit in constant-width decomposition
 Therefore: either violates thickness bound ($k > 12$) or requires super-polynomial resources
 Contradiction. ■

Formal statement:

SAT $\notin$ (Polytime $\cap$ Bounded-Separator-Computation)

4. Open Question: Model Equivalence

Central question: Does $P =$ Bounded-Separator-Polytime?

Two scenarios:

Scenario A: If $P = k$ -Rail-Admissible (for $k = \text{polylog}(n)$), then Theorem A implies **$P \neq NP$** .

Scenario B: If $P \supsetneq k$ -Rail-Admissible, we've defined an interesting proper subclass of P (possibly related to NC, SC, or bounded-treewidth circuits) and shown SAT is outside it.

Current status: Model Equivalence unproven. Most likely Scenario B: polynomial-time algorithms can use polynomial space (large separators), placing them outside the rail model.

Value regardless: The separation is interesting because:

Uses geometric constraints (treewidth) rather than resource bounds
Connects graph structure to computational limits
Identifies structural barriers independent of traditional complexity measures
Provides new perspective on why certain problems are hard

5. Request for Expert Assessment

I seek feedback on:

Correctness: Are Lemmas 1-3 proven correctly? Any gaps?

Novelty: Is rail-admissible equivalent to known models (bounded-treewidth circuits, NC, SC)?

Model Equivalence: Can P be shown equal to $f(n)$ -rail-admissible for $f(n)=\text{polylog}(n)$? What's the obstruction?

Significance: Is Theorem A interesting independent of P vs NP?

Related work: What should I cite? Similar separations via treewidth?

A technical supplement with full proofs is attached.

References

- [1] Ben-Sasson, E., & Wigderson, A. (2001). Short proofs are narrow—resolution made simple. *Journal of the ACM*, 48(2), 149-169.
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